BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 091 · High-Level and Low-Level Languages

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Evaluate (A AND B) OR C for A=1, B=0, C=1.

Q2. What is denary 182 in unsigned 8-bit binary?
A. 10101110
B. 11010110
C. 10110110
D. 01101101

Q3. Add 00010111 and 00000110 as unsigned binary numbers. Give the eight-bit result.

Q4. (a) Logically shift unsigned eight-bit 00110001 one place left. Give the eight-bit result and denary value. (b) Logically shift unsigned eight-bit 00110001 one place right. Give the eight-bit result and denary value.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

High-level language — A programming language with readable instructions less tied to a particular CPU.

Low-level language — A language close to machine instructions, such as assembly language or machine code.

Machine code — Binary instructions the CPU can execute.

Portability — How easily a program can be moved to different hardware or systems.

Translator — Software that converts source code into a form the computer can execute.

New learning

High-level languages: High-level code is designed to be readable: for example, score = score + 1. It is usually easier to develop, debug and maintain than processor-specific instructions.

Low-level languages: Machine code is made of binary instructions executed by a CPU. Assembly is a low-level form using short instruction mnemonics: you only need to recognise the difference, not learn an assembler.

Why translate?: Processors execute machine code. Source programs written in high-level languages need translation into executable instructions before or while running.

Choose for the task: A school application benefits from readable, maintainable, portable code. Direct processor control may justify a lower-level approach, but is harder to maintain and is processor-dependent.

Worked example

A school is developing a timetable application. Explain TWO benefits of using a high-level language. [4 marks]

1. The application will be maintained and may be used on different computers. Choose benefits relevant to that context.

2. High-level code is more readable [1], so staff can understand and change the program more quickly [1].

3. High-level code is generally more portable [1], so moving to a different processor is likely to require less rewriting [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

4. Award a knowledge mark and a linked consequence for EACH explained benefit. Do not simply name two features.

We Do · Practise with guidance

Guided 1

A controller needs precise access to processor instructions. Explain ONE advantage and ONE disadvantage of using a low-level language. [4 marks]

Worked method: Instructions can give precise processor control [1] useful for managing the device hardware [1]. They are processor-specific [1], so code may need rewriting for different hardware [1].

Guided 2

Compare high-level with low-level code in terms of readability and portability. [4 marks]

Worked method: High-level programs are usually easier for people to read [1] and are more portable [1]. Low-level code corresponds closely to CPU instructions [1] and is usually processor-specific [1].

Guided 3

A school wants an attendance application that teachers can edit and run on different computers. Recommend a language level and explain TWO reasons. [4 marks]

Worked method: High-level [1]; the code is easier to read and maintain as rules change [1]; it is usually more portable [1], making it easier to move across processor families [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write one OCR ERL line to print the character represented by ASCII code 66.

```
code = 66
```

write

Question 1

A new colleague takes over a school reports program. Explain TWO advantages of the code having been written in a high-level language. [4 marks] Source: Original OCR J277-style practice question.

Question 2

A computer is upgraded to a processor with a different instruction set. Explain TWO possible drawbacks of an existing low-level program. [4 marks] Source: Original OCR J277-style practice question.

Question 3

A medical practice is creating an appointment-booking program that will need regular updates. Explain TWO advantages of using a high-level language. [4 marks] Source: Original OCR J277-style practice question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

Machine code is made of processor instructions that the CPU executes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 1: A AND B is 0, then 0 OR 1 is 1.

Q2. C — 10110110

Q3. 00011101; 23 + 6 = 29.

Q4. (a) 01100010 = 98. Fill the empty bit with 0. (b) 00011000 = 24. Fill with 0 and lose the old rightmost bit.

I Do / We Do pages · Guided practice

Q1. Instructions can give precise processor control [1] useful for managing the device hardware [1]. They are processor-specific [1], so code may need rewriting for different hardware [1].

Q2. High-level programs are usually easier for people to read [1] and are more portable [1]. Low-level code corresponds closely to CPU instructions [1] and is usually processor-specific [1].

Q3. High-level [1]; the code is easier to read and maintain as rules change [1]; it is usually more portable [1], making it easier to move across processor families [1].

You Do pages · Code First

```
code = 66
print(CHR(code))
```

Expected result: B

You Do pages · Independent questions

Q1. High-level code is easier to read [1], helping the colleague understand existing instructions [1]. It is generally easier to maintain [1], making changes and fault correction quicker [1].

Q2. Low-level instructions are often specific to the original processor [1], so the new processor might not execute the code [1]. The program may need rewriting for the new instruction set [1], increasing development time and maintenance costs [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Q3. Readable code makes maintenance and correction easier [1], saving development time when appointment rules change [1]. High-level code is generally portable [1], reducing rewriting if the system is moved to other processor types [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: